From Scalar Charge to Local Force

Accuracy, Distance Dependence, and Threshold Decisions in a Spherical Chameleon Field

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Abstract

A screened source is often replaced by one effective scalar charge when predicting a fifth force. This study tests two separate steps: estimating that asymptotic charge and using it to predict the force at a finite distance. For $V(\phi) = \Lambda^5/\phi$ in the weak conformal-coupling limit, a two-domain collocation solver computes 725 nonlinear spherical boundary-value problems with $10^2 \leq \rho_c/\rho_\infty \leq 10^5$ and $10^{-3} \leq m_\infty R \leq 0.3$. An active-shell Yukawa integral predicts the charge within 10% at 99.7% of this grid. A simpler point-factor times finite-volume control reaches the same coverage; its median error is 0.378%, versus 0.391% for the radial integral. The additional radial weighting is therefore not established as the cause of the improvement. Even the numerical asymptotic charge underpredicts the local force at $1.5R$ by as much as 16.6%; 100 grid points exceed 10% error. The maximum errors fall to 3.85% at $5R$ and 1.84% at $10R$. At the illustrative force-ratio threshold $\alpha_\star = 10^{-2}$, replacing the tail-based reference with the direct nonlinear gradient changes 2 decisions. A refined-domain audit of 16 selected points preserves the direct-force decisions. These are numerical validity and decision-boundary results for an idealized model, rather than detector sensitivities or new experimental exclusions.

1 Research question and scope

Chameleon screening changes the strength and range of a scalar-mediated force through the ambient matter density [1, 2]. A dense body can have a nearly constant interior field and a small effective scalar charge, despite an appreciable microscopic matter coupling. The familiar long-range thin-shell estimate provides a useful, inexpensive description of that effect. Its use in an observation, however, involves both a source approximation and a force-propagation approximation.

The questions here are: (i) how accurately do specified charge models reproduce the nonlinear spherical solution, (ii) how much of the improvement comes from finite shell volume rather than radial Yukawa weighting, and (iii) how far from the source must a probe be before the measured asymptotic charge predicts its local force accurately? The last question supplies a direct mechanics connection: the acceleration of a small test body is set by the local field gradient, not by an asymptotic coefficient alone.

Analytic thin-shell and no-shell matching, including finite interior-mass effects, already exists [3]. Real laboratory geometries have also been solved numerically [7, 8]. This study makes no priority claim for these methods or for Yukawa shell integration. Its limited contribution is a reproducible charge-to-local-force error analysis with a controlled ablation and an explicit threshold reference. All sources and data are available in the public study repository.

2 Physical model and normalization

Use natural units $\hbar = c = 1$ and the reduced Planck mass $M_{\rm Pl} = (8\pi G)^{-1/2}$. For a positive scalar field take

$$V(\phi) = \frac{\Lambda^5}{\phi}, \qquad A(\phi) = e^{\phi/M_c}, \qquad M_c = \frac{M_{\rm Pl}}{\beta}. \tag{1}$$

The static weak-gravity equation is

$$\nabla^2\phi = -\frac{\Lambda^5}{\phi^2} + \frac{\rho}{M_c}e^{\phi/M_c}. \tag{2}$$

To make the approximation explicit, define $x = r/R$, $u = \phi/\phi_\infty$, $\widetilde{\rho} = \rho/\rho_\infty$, and $D = \phi_\infty/M_c$. The full dimensionless equation has coefficients

$$u'' + \frac{2}{x}u' = -\frac{A_0}{u^2} + B_0\widetilde{\rho}e^{Du}, \quad A_0 = \frac{\Lambda^5 R^2}{\phi_\infty^3}, \quad B_0 = \frac{\rho_\infty R^2}{M_c\phi_\infty}. \tag{3}$$

At the exact ambient minimum $A_0 = B_0 e^D$ and $(m_\infty R)^2 = A_0(2+D)$. The present experiment deliberately takes $D \to 0$; it does not silently discard an independent coupling parameter. In that limit

$$\phi_{\rm min} = \left(\frac{\Lambda^5 M_c}{\rho}\right)^{1/2}, \qquad m_{\rm eff}^2 = \frac{2\Lambda^5}{\phi_{\rm min}^3}, \qquad \lambda_C = m_{\rm eff}^{-1}. \tag{4}$$

The two inputs are $\delta = \rho_c/\rho_\infty$ and $\mu = m_\infty R$:

$$u'' + \frac{2}{x}u' = \frac{\mu^2}{2}[\widetilde{\rho}(x) - u^{-2}], \quad \widetilde{\rho} = \begin{cases}\delta, & x < 1,\\ 1, & x > 1,\end{cases} \quad u'(0) = 0, \quad u(\infty) = 1. \tag{5}$$

Consequently $u_c = \delta^{-1/2}$ and $m_c R = \mu\delta^{3/4}$.

2.1 A parameter is not a measured geometric thickness

The normalization uses the total body mass $M_s = 4\pi\rho_c R^3/3$ and $\Phi_N = GM_s/R$. Define

$$\epsilon_{\rm th} \equiv \frac{\phi_\infty - \phi_c}{6\beta M_{\rm Pl}\Phi_N} = \frac{2(1-\delta^{-1/2})}{\mu^2\delta}. \tag{6}$$

This is the potential-defined thin-shell parameter, not an exact geometric thickness. In the relevant large-interior-mass asymptotics, $\epsilon_{\rm th} \simeq \Delta R/R + 1/(m_c R)$; identifying it with $\Delta R/R$ requires the latter correction to be negligible [3]. For example, at $(\delta, \mu) = (10^5, 0.1)$ the parameter is 1.99×10^{-3} while $1/(m_c R) = 1.78 \times 10^{-3}$. We do not label an assumed shell radius as a thickness measured from the BVP.

3 Charge models and the ablation control

All charge predictions below use the same surface-referenced tail convention. The clipped leading long-range law is

$$Q_{\rm LR} = \min(3\epsilon_{\rm th}, 1), \qquad \mu \ll 1. \tag{7}$$

Clipping at one is already an interpolation through the transition, rather than a controlled expansion there. Two finite-range amplitude hypotheses are

$$Q_{\rm surf} = Q_{\rm LR}, \qquad Q_{\rm ctr} = e^{-\mu}Q_{\rm LR}. \tag{8}$$

These are different approximations to the same coefficient. They are not a physical ambiguity in the theory. For a fixed solution, changing from $e^{-\mu(x-1)}/x$ to $e^{-\mu x}/x$ simply changes the

coefficient by $Q^{\text{center reference}} = e^{\mu} Q^{\text{surface reference}}$ and leaves the field and force invariant. Nor does "surface amplitude" mean the exact field of a material spherical shell. Finite-mass matching supplies more detailed approximations than these two simple hypotheses [3].

A massless matching model with a frozen core gives the assumed radius fraction $q = \sqrt{\max(1 - 2\epsilon_{\text{th}}, 0)}$ and finite active volume

$$Q_{\text{MS}} = 1 - q^3. \tag{9}$$

This construction is an approximation; the finite-μ core is not rematched self-consistently. To isolate the extra radial weighting, compare

$$Q_{\text{PV}} = e^{-\mu}(1 - q^3), \tag{10}$$

$$Q_{\text{AS}} = \frac{3e^{-\mu}}{\mu^3}[g(\mu) - g(\mu q)], \qquad g(z) = z\cosh z - \sinh z. \tag{11}$$

Both have the same assumed active volume and point Yukawa factor; only Q_{AS} includes the additional spherical radial weight. In the fully active limit the exact linear uniform-sphere result also contains the background subtraction $1 - \delta^{-1}$:

$$Q_{\text{lin}} = (1 - \delta^{-1})\frac{3e^{-\mu}}{\mu^3} g(\mu). \tag{12}$$

The shell models omit this small background term on the high-contrast grid. It is retained in the weak-source verification.

4 Asymptotic charge and local acceleration

In the linear exterior region define the numerical charge by

$$u = 1 - sQ_{\text{num}}\frac{e^{-\mu(x-1)}}{x}, \qquad s = \frac{\mu^2\delta}{6}. \tag{13}$$

Two estimators, used only in a suitable tail interval, are

$$Q_A = \frac{x(1-u)e^{\mu(x-1)}}{s}, \qquad Q_\nabla = \frac{x^2 u' e^{\mu(x-1)}}{s(1+\mu x)}. \tag{14}$$

The reported Q_{num} is the median of Q_A where the nonlinear exterior source and its linearization differ by less than 2%. Agreement with Q_∇ checks tail extraction; it does not establish linearity near the source.

For a small, non-backreacting probe with a supplied response factor Q_t, the nonlinear reference and the Yukawa-tail prediction are respectively

$$\alpha_{\text{direct}}(x) = \left|\frac{a_5}{a_N}\right|_{\text{direct}} = \frac{2\beta^2 Q_t}{s} x^2 u'_{\text{num}}(x), \tag{15}$$

$$\alpha_{\text{tail}}(Q, x) = 2\beta^2 Q_t Q(1+\mu x)e^{-\mu(x-1)}. \tag{16}$$

Here $a_N = GM_s/(R^2x^2)$ is the body's Newtonian reference, not the acceleration of an infinite uniform environment. The main force products use $\beta = Q_t = 1$. The factor Q_t is not a numerical solution of a two-body screening problem.

There are two distinct multiplicative errors:

$$\frac{\alpha_{\text{tail}}(Q_j, x)}{\alpha_{\text{direct}}(x)} = \frac{Q_j}{Q_{\text{num}}}\frac{\alpha_{\text{tail}}(Q_{\text{num}}, x)}{\alpha_{\text{direct}}(x)}. \tag{17}$$

Thus charge error and propagation error can reinforce or cancel each other. A charge-accuracy ranking need not be a local-force-accuracy ranking.

5 Numerical method and verification

The density jump at $x = 1$ is preserved with separate inner and outer collocation domains mapped to a shared coordinate. The solver matches both u and u' at the surface. At the finite outer boundary it imposes the asymptotic Robin condition $u'(x_{\max}) = (\mu+1/x_{\max})[1-u(x_{\max})]$. The charge-scaled variables $x(u-1) = sv$, or $u = \exp(sv/x)$ when positivity is helpful, improve conditioning. Both represent the same nonlinear equation; they are not thin-shell inputs to the BVP. The Robin condition is asymptotic: in the logarithmic representation it is imposed on $x \log u/s$ to the same linear-tail order. The domain audit checks the effect of the remaining nonlinear boundary terms.

SciPy's fourth-order collocation algorithm solves a 25×29 logarithmic grid over the ranges in the abstract, nominally at $x_{\max} = 160$ with tolerance 2×10^{-4}. Every point converges. The solution is sampled directly at 71 probe radii in $1.02 \le x \le 20$, including $1.05, 1.5, 2, 3, 5, 10$. The weak-source uniform-sphere benchmark has maximum relative charge error 2.7×10^{-5}.

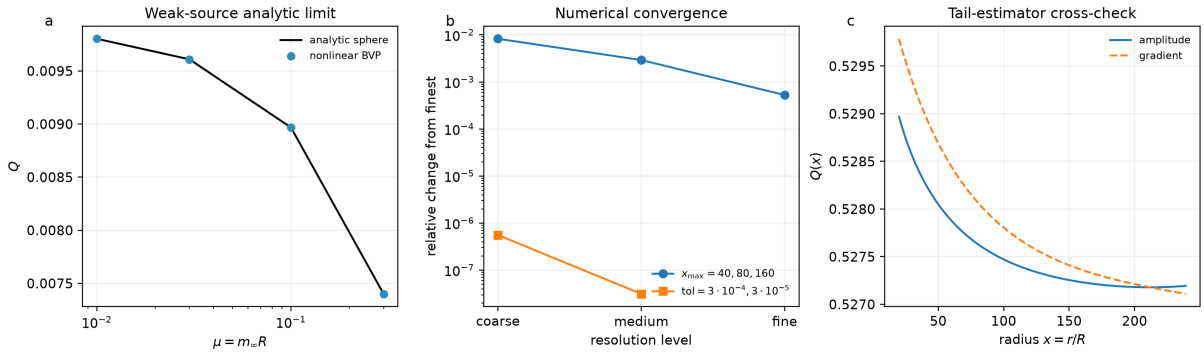


Figure 1: Analytic-limit, resolution, and tail-estimator checks. Agreement of the two tail estimators is a diagnostic within one BVP solution, not a second independent field solver.

The refined audit selects corners, representative points, points near the charge-model 10% boundaries, the tail/direct threshold flips, and the largest local-force error. At 16 points it compares $160R$ and $240R$ with tightened tolerances. The largest charge change is 0.056%. Across the sampled probe radii the largest local-force difference between the scan and refined solution is 0.0026%; the direct threshold classification changes at 0 audited points. This supports the reported contrasts but is not a convergence certificate for every grid point or a continuous parameter-space bound.

6 Results

6.1 Charge accuracy and what the ablation establishes

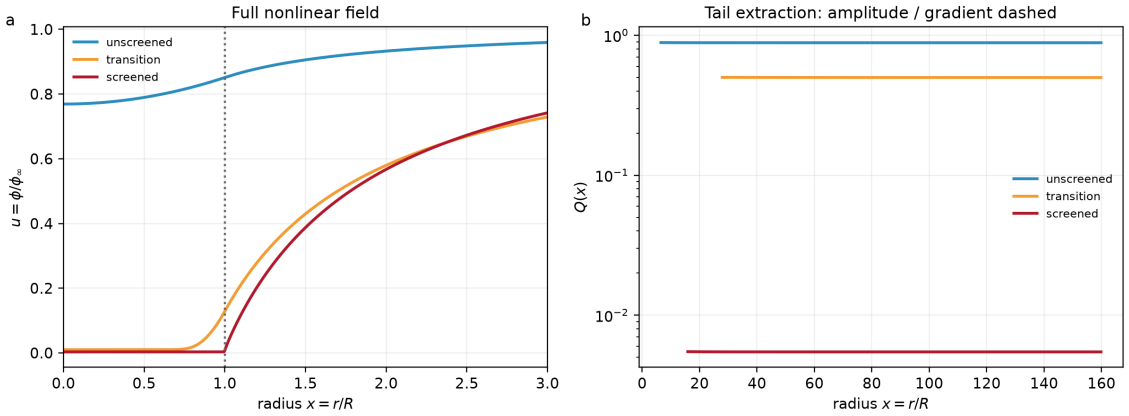


Figure 2: Representative unscreened, transitional, and strongly screened solutions. The charge plateau is extracted from the exterior solution.

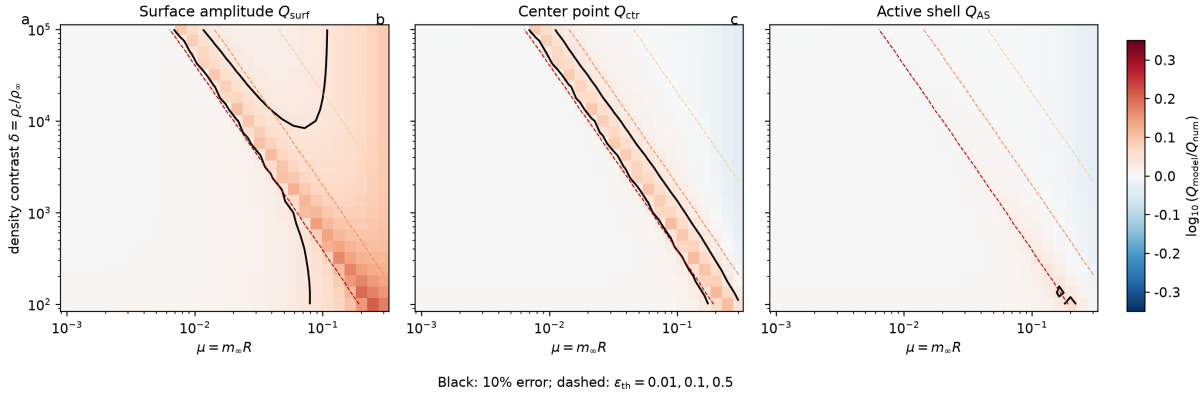


Figure 3: Signed charge errors against the numerical asymptotic charge. Black contours mark 10% error. The first two panels compare finite-range amplitude hypotheses in one common normalization.

Table 1: Charge errors on the logarithmic grid. Coverage counts are not probabilities.

Model	Median error	Maximum	Within 10%
Surface-amplitude model	2.78%	62.6%	499/725
Center-point model	0.566%	30.1%	662/725
Finite shell volume	2.18%	37.6%	562/725
Point + finite volume	0.378%	12.7%	723/725
Active-shell integral	0.391%	13.1%	723/725

The simpler Q_{PV} and the radial integral both reach 99.7% coverage. The former has the smaller median and maximum errors, whereas the integral has a slightly smaller 90th-percentile error (recorded in the data). The extra radial weight changes Q_{PV} by at most 1.51% on this grid. The evidence supports an effective combined finite-volume/Yukawa correction; it does not identify radial weighting as necessary for the observed improvement.

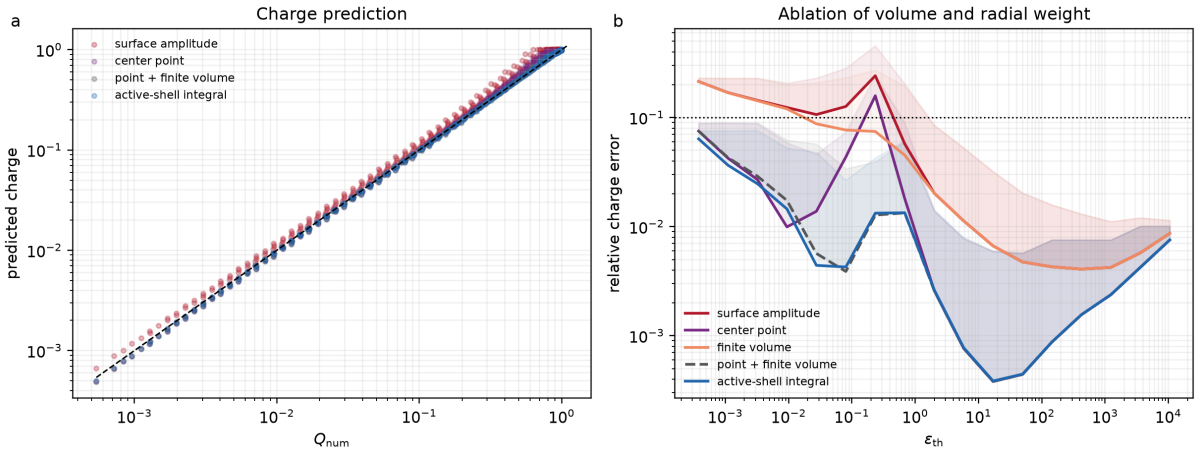


Figure 4: Ablation control added to the charge comparison. Dashed gray is the point-factor plus finite-volume model; shaded bands run from median to 90th-percentile error within bins of ϵ_{th}.

The empirical long-range charge filter $\epsilon_{\mathrm{th}} < 0.1$, $m_c R > 10$, $\mu < 0.035$ passes 17 grid points (2.34% coverage) and 12 held-out log-midpoints at 10% accuracy. It is a narrow empirical check, not a universal sufficient condition. On this high-contrast grid, $\epsilon_{\mathrm{th}}(m_c R)^2 = 2(\sqrt{\delta} - 1)$ makes $m_c R > 10$ redundant once $\epsilon_{\mathrm{th}} < 0.1$ holds; the scan cannot independently establish the role of that cut.

6.2 Distance-dependent validity of the tail force

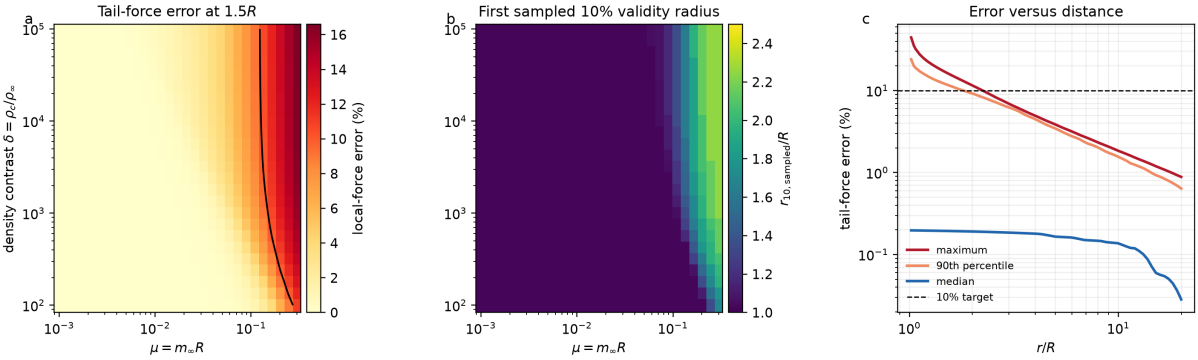


Figure 5: (a) Force error at $1.5R$ using the numerical charge itself. (b) First sampled radius satisfying 10% error at every later sampled radius through $20R$. (c) Error statistics versus radius. The sampled boundary is not a proof of continuous validity between samples or beyond $20R$.

Table 2: Numerical-charge tail force versus the direct nonlinear gradient.

r/R	Median error	Maximum error	Points above 10%
1.05	0.197%	35.4%	148
1.5	0.194%	16.6%	100
2	0.191%	11.6%	52
5	0.165%	3.85%	0
10	0.137%	1.84%	0

Even exact knowledge of the asymptotic charge does not remove the near-source propagation error. At $1.5R$, 100 points exceed 10% and the largest error is 16.6%. On the sampled radial grid all points have reached the 10% target by $r/R = 2.25$, with all later samples also passing. The fixed $5R$ and $10R$ slices are stronger empirical checks at the corresponding sampled radii, with maximum errors of 3.85% and 1.84%.

Table 3: The same charge models used to predict the local force at $1.5R$.

Model	Median error	Maximum	Within 10%
Surface-amplitude model	1.37%	48.1%	629/725
Center-point model	0.474%	24%	594/725
Finite shell volume	0.941%	28.8%	690/725
Point + finite volume	0.423%	24%	616/725
Active-shell integral	0.424%	22.9%	618/725

For example, Q_{AS} gives 99.7% charge coverage but 85.2% local-force coverage at this radius, with maximum local error 22.9%. This change follows the error factorization in equation (17); a favorable charge benchmark alone is insufficient.

6.3 Threshold decisions use the direct force

The map now uses equation (15) at $1.5R$ as its numerical reference and $\alpha_\star = 10^{-2}$ as an illustrative threshold. The tail computed from Q_{num} disagrees with that reference at 2 points. The surface-amplitude and center-point hypotheses disagree with the direct result at 1 and 3 points, respectively. Their median contour shifts are 0.00156 and 0.0369 dex; these are log-grid interpolations, not resolved continuous boundaries. Their mutual decision disagreement is 4 points and both are wrong at 0 points for this threshold.

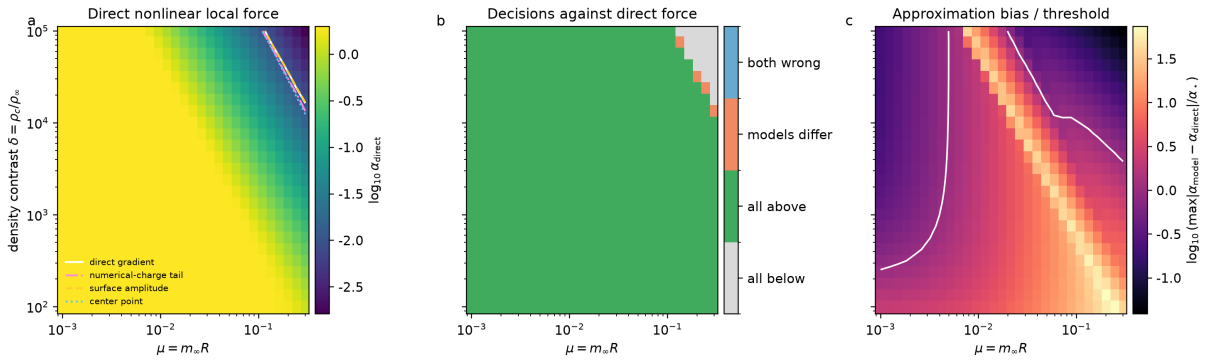


Figure 6: Local force-ratio threshold map. White is the direct-gradient contour; the numerical-charge tail and two simple amplitude predictions are shown separately. The third panel normalizes their bias by the selected threshold.

The separate sensitivity table spans $\alpha_\star = 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1$. All-above and all-below cases are legitimate outputs: if no contour crosses the grid, contour displacement is recorded as null rather than causing a reduction of an empty array. No claim about arbitrary thresholds follows from one map.

7 Interpretation, limitations, and reproduction

The supported research result is a distinction between charge error and finite-distance acceleration error, with a directly tested distance dependence. The ablation also shows that the more

elaborate radial integral is not required to obtain its charge coverage on this restricted grid. Whether that changes for $\mu \gtrsim 1$, other potentials, or different source shapes remains open; such extensions require new numerical checks and a broader comparison with existing matched solutions.

The model is static, spherical, homogeneous in each region, nonrelativistic, $n = 1$, and conditional on $D \to 0$. A real experiment additionally needs the source's dimensional scale, chamber walls, finite test-body response, source modulation, detector noise and systematic errors. Existing atom and levitated force searches illustrate the importance of those ingredients [6, 9, 10]. This isolated-sphere map supplies no new physical (Λ, M_c) exclusion and does not reassess those experiments. For an actual forecast the relevant quantity is a predicted differential signal divided by its uncertainty, not an arbitrary force-ratio cut.

From the repository root, run:

```
python -m Thin_Shell_Study.run_study --workers 4 --force
python -m unittest discover -v
```

The first command generates both languages' figures, scan and local-force CSVs, distance and threshold-sensitivity tables, refined audits, JSON metrics, and the numeric TeX inputs used by both reports and decks. Cached data carry a schema and numerical-model/settings fingerprint; a charge-only or stale cache is recomputed. The JSON records the package versions, source fingerprint, and Git revision at the run. The Korean-only rendering command reads the same metrics and threshold, with no independent numerical defaults.

8 Conclusion

The numerical charge benchmark remains reproducible, but it is not by itself a force benchmark near the source. The point-factor/finite-volume control matches the active-shell integral's charge coverage, limiting the causal claim for radial weighting. The direct-gradient comparison exposes a 16.6% worst tail-force error at $1.5R$, decreasing to 3.85% at $5R$ and 1.84% at $10R$ on this grid. Using that local force changes 2 decisions at the selected threshold. These results provide a concrete way to control the approximation used between a screened source and a classical-mechanics observable.

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